The Problem

- Using a **simple retention model (Retention Part 3)**, customer lifetime value (**LTV**) is:

$$LTV = \sum_{t=1}^{\infty} \frac{m_t \boxed{r}^{t-1}}{(1+\delta)^{t-1}}$$

  - m_t: customer's profit contribution in period t
  - r: retention rate   Focus of churn management
  - δ: discount rate

- **Definition** of churn:
  - **Customer level**: probability the customer leaves in a given period
  - **Firm level**: proportion of customer base that leaves in a given period

> ➢ Churn $= c = 1 - retention\ rate = 1 - r$

The Problem

Two types of churn

1. **Involuntary** churn
   - **Company decides** to terminate relationship with customer
2. **Voluntary** churn
   - **Customer decides** to terminate relationship with company
     a) **Deliberate voluntary** churn
        - Customer dissatisfied
        - Customer received better offer

     b) **Incidental voluntary** churn
        - Customer no longer needs product
        - Customer moved to location not serviced by company

The Problem

- Table 24.1 (Blattberg et al., p. 609): **20% - 50%** of customers that start the year with a company, **leave** by year end
- **To illustrate** importance of such churn rates:

1. **Expected customer lifetime**
   - ➢ How long customer stays with company
   - ➢ $= \frac{1}{c}$, reciprocal of churn rate (see Appendix 24.1, Blattberg et al.)

Figure 24.1, p. 610, Blattberg et al.

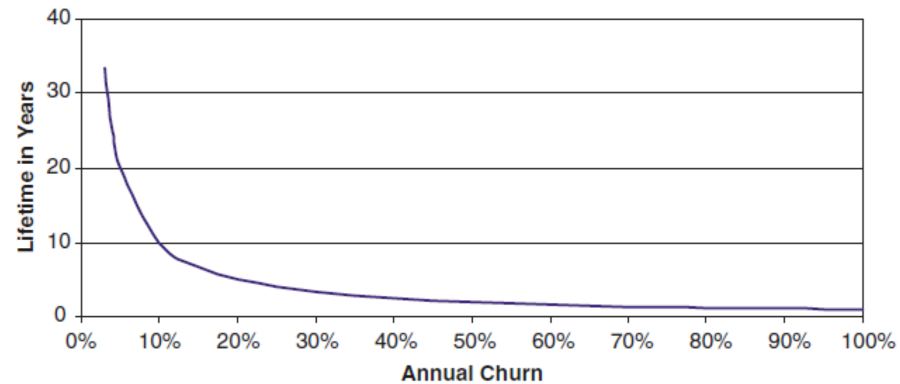


Fig. 24.1 Expected lifetime as function of churn rate (Equation 24.3).

Annual churn rate	50%	40%	30%	20%
Expected lifetime (years)	2	2.5	3.3	5

➢ Lower churn rate, longer lifetime

The Problem

- Lower churn rate, higher lifetime value
- Lower churn rate, higher ROI

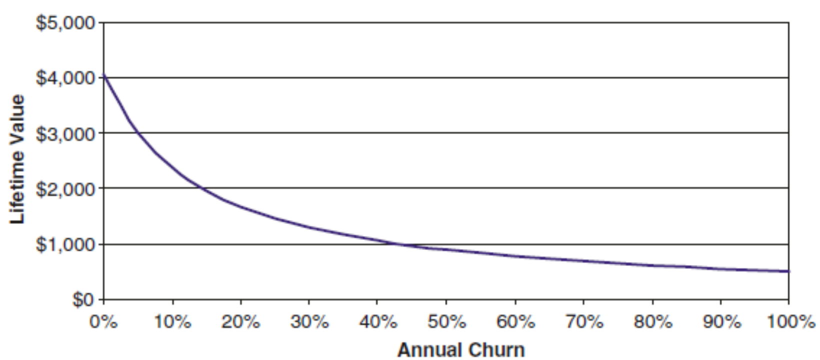
2. <u>Customer lifetime value</u>

$$LTV = \sum_{t=1}^{\infty} \frac{m_t \boxed{r}^{t-1}}{(1+\delta)^{t-1}}$$

$$LTV = \sum_{t=1}^{\infty} \frac{m\boxed{(1-c)}^{t-1}}{(1+\delta)^{t-1}} = \boxed{\frac{m(1+\delta)}{(\delta+c)}}$$

- m: annual profit contribution per customer
- c: annual churn rate
- δ: annual discount rate

<u>Figure 24.2</u>, p. 610, Blattberg et al. (Assuming $\delta = 14\%$ & $m = \$500$ per year)



Annual churn rate	50%	40%	30%	20%
Lifetime value ($ per customer)	891	1056	1295	1676
ROI (acq cost = $400)	123%	164%	224%	319%

Factors that Cause Churn

- The first step towards predicting and remedying churn is to **understand the factors** that cause or alleviate it

CUSTOMER SATISFACTION	SWITCHING COSTS	CUSTOMER CHARACTERISTICS
• Service quality • Fit-to-needs • Meeting expectations • Price	• Physical • Psychological	• Risk aversion • Variety seeking • Deal proneness • Mavenism

MARKETING	COMPETITION
• Reward programs • Promotions • Price • Customized products	• Within category • Between category

See p. 611-615 in Blattberg et al. for discussion of other factors

Switching Costs

Low switching cost ⇒ Increased churn probability

<u>Two forms</u>:

1. Physical switching cost
   - Real inconvenience
   - E.g. "mobile number portability"
2. Psychological switching cost
   - Inertia
   - Brand pull
   - Familiarity

- **<u>Interesting example</u>**: number portability

Customer Characteristics

Higher churn likelihood

- Risk takers
- Variety seekers
- Shopping mavens
- Deal prone customers
- Lower income customers

Lower churn likelihood

- Rural customers
- Higher income customers
- Married customers

Marketing Efforts

Effective marketing ⇒ Decrease churn

✓ Things found to **work**:

- Special services
- Loyalty programs
- Matching customers to the right price plans

Proactive Churn Management

- **Key**: identifying customers likely to churn
  - **Predictive modelling**

- No data on causes of churn
- No psychographic data (risk-taking)
- Competitive activity

- Behavioural measures (R, F, M)
- Customer complaints
- Offers extended to the customer
- Price information
- Any retention efforts
- Etc.

Single Future Period Models in Action: 1

The "Balance Transfer Flight Risk" Profile

- **<u>From predictors</u>:** High *Interest charged month 3* + High *Credit card balance month 6*.
- This customer is carrying a heavy balance and getting hit with high interest charges. They are highly susceptible to a competitor's "0% APR on Balance Transfers" promotional offer.
  - APR: annual percentage rate
  - Total annual cost of borrowing money, expressed as a percentage, including the interest rate, fees, and other charges

➢ **<u>Possible incentive</u>**: Proactively offer them a temporarily lowered APR for the next 6 months.

   ➢ Offer to convert their current high-interest balance into a fixed-fee, lower-interest installment plan. This retains the debt within your bank but lowers their monthly burden, removing the incentive to switch.

Single Future Period Models in Action: 1

"Balance Transfer Flight Risk" ⇒ Cost of False Negatives?

- **False negative**: the model predicts the customer will stay, so the bank does nothing, but the customer actually churns
- For a customer with a **high balance**, their expected future value **(CLV) is high** because they generate significant monthly interest revenue for the bank on this high balance.
- If the bank uses a default threshold like 0.5 to turn probabilities into churn predictions that would trigger interventions on their system, the model will probably generate too many false negatives. The bank would miss the chance to save a high CLV customer, taking a huge financial hit.
  - ➢ **Lower the threshold**, deliberately accepting more false positives (offering lower rates to customers who might have stayed anyway) just to ensure they do not commit a costly false negative.

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Single Future Period Models in Action: 2

The "Lost Wallet Share" Profile

- **From predictors:** Decrease in *Purchase level month 6 + Household age* demographic.
- The customer isn't necessarily closing the account right away, but they have stopped using your card for daily spending. They have likely moved a competitor's card to the "top of their wallet" because of better rewards tailored to their age/lifestyle.

➢ **Possible incentive**: Spend and Get Promotion.

  ➢ Offer a targeted, short-term incentive, such as "Earn 5,000 bonus points when you spend R2,000 on groceries this month." This forces them to pull your card back out of their wallet and re-establish the habit of using it.

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Single Future Period Models in Action: 2

“Lost Wallet Share" ⇒ Threshold adjustment?

- The **threshold** for action is directly proportional to how much the intervention costs.
- The proactive measure here – spend R2 000 and get 5 000 points – is very cost efficient for the bank. They will only incur a cost if the customer spends the money.
- Every time a customer swipes a credit card, the bank collects an **interchange fee** from the merchant (usually between 1.5% and 3% of the transaction). If the promotion requires the customer to spend R2,000 on groceries to get the bonus, the bank is actively generating R30 to R60 in new revenue. This interchange revenue offsets the cost of the reward points they are giving away.
- Because the cost is so low the bank can afford to give a promotion to a customer that wasn’t really going to churn. They can thus **accept more false positives**
  - **Lower the threshold**

Time Series Models

Chapter 6 Kumar & Petersen (2012)

Why and when is a customer likely to churn?

- **Example**: B2B firm, contractual basis
  - Churn observed
- **Modelling** focuses on whether or not customer defected in the **1st two years (730 days)** of the relationship
  - **Non-defects**: observe **censored lifetime** of 730 days
- **Aim**: understand difference between customers that have churned and customers that haven't left yet:
  - Determine **drivers of customer churn**
  - **Predict expected duration** of customers yet to churn
  - Determine **predictive accuracy** of model

The AFT Model

- **Parametric** alternative to Cox Proportional Hazard
- Examines predictors' effects on event times in **censored data** regression
  - Looks at 1st two years of relationship (730 days)
- **Model**:

$$\ln(Duration_i) = X_i'\beta + \sigma\varepsilon_i$$

  - $\ln(Duration_i)$: natural log of customer i duration (**log-normal dist**)
    - **Other** possible distributions: Exponential, Weibull, log-logistic
  - X_i: predictors for customer i
  - β: coefficients
  - σ: scale parameter
  - ε_i: random error

INCENTIVES TO PREVENT CHURN

Managerial Approaches to Reducing Churn

<u>**Untargeted**</u> approaches

- Increase customer satisfaction / switching costs
- Improved products, advertising or loyalty programs

<u>**Targeted**</u> approaches

- Try to rescue customers identified as likely to churn

Managerial Approaches to Reducing Churn

Two types of **targeted approaches**

- **Reactive**
  - Wait for customers to identify themselves as likely to churn
  - E.g. calls to cancel service
- **Proactive**
  - Advanced identification of likely churners
  - Diagnose reason for potential churn
  - Takes appropriate action to prevent churn

Reactive Churn Management

Perfect/near-perfect prediction of churn

- Always addressing the correct customers with regards to churn management
- Company **can afford significant incentive** to prevent churn

Large incentive costs a lot in the short term

- Can be **costly due to the size** of the incentive needed to convince a leaving customer to stay
- May **incentivise** customer to call whenever customer receives competitive offer

Proactive Churn Management

Relies on a predictive model to flag possible churners

- Predictive accuracy depends on model quality
- Predictions often poor due to lack of data
- Could be missing out on profitable customers leaving the business without your knowledge

Might **stimulate satisfied customers to consider churning**

- Company **cannot spend too much** per incentive to limit potential waste since you **might make an offer to a customer that wasn't going to churn**

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Framework for Proactive Churn Management

- Potential **trade-offs** to proactive churn management:
  - **Predictive accuracy** of model
  - **Effectiveness** of **offers** made
  - **Potential wastage** of offers made to non-churners

- These issues can be quantified by a **profitability framework**
- Profitability framework can be used in two ways:
  1. Will a **proposed offer** result in some **increase** in **revenue**?
  2. **How much should be offered** if the business seeks **x% increase** in **ROI**?